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GLOBAL_MULTIYEAR_BGC_001_033

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table. The third column specifies which CMEMS-GLO-QUID-001_033-V4.2_Revision_v2.0_ABsections, or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

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4.2	June 2024	All	Change of computation grid	Olivier Titaud Anna Conchon Laurène Mérillet Nicolas Goeman Héloïse Magliano	Olivier Le Galloudec

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EXECUTIVE SUMMARY

I.1 Products covered by this document

GLOBAL_MULTIYEAR_BGC_001_033: The global low and mid-trophic level (LMTL) daily hindcast is delivered on a standard latitude-longitude horizontal and regular 1/12° resolution grid. Low trophic level (LTL) is represented by the density of meso-zooplankton. Mid-trophic level (MTL) is represented by six functional groups of micronekton. Traditionally, micronekton is defined by a size between 2cm and 20cm and it contains a large diversity of fish, crustaceans, squid, and gelatinous species. For species with large growth potentials, it represents a temporary stage.

This product uses a simplified 3-layer vertical structure to represent the pelagic habitats of the organisms considered here (see section II.1.1) and delivers the bottom depths of each layer as output. We also present some physical environment variables (temperature and current velocities) depth-averaged over these pelagic layers.

This product also bears the name of MICRORYS (MICRORYS12V5) within the GLO-HR MFC for correspondence to other global multi-year products like GLORYS, BIORYS, etc. This document is associated with the version of the product currently available in the Copernicus Marine Service online catalogue[1].

I.2 Summary of the results

The quality of the global daily LMTL hindcast simulation GLOBAL_MULTIYEAR_BGC_001_033 from 01/01/1998 to 31/12/2023 has been assessed through a comparison with existing observations (zooplankton) in publicly accessible databases[2], information from the literature (micronekton, see Table 7 and associated bibliography in section VII) and observations based on estimates from acoustic transects (pelagic layer depths, see section II.3). The results are summarized below.

The GLOBAL_MULTIYEAR_BGC_001_033 simulation product uses biogeochemical and physical products as forcing terms. Hence, the quality of this product strongly depends on the quality of this input forcing, especially on net primary productivity (NPP). NPP is computed from satellite observations, complemented by estimates from a biogeochemical model when observations are not available. Temperature and currents come from an ocean global circulation model (OGCM) reanalysis. See more details about the used products in Sections II.2 and II.3.

While the product is provided for latitudes up to 80°S-90°N, results at high latitudes (i.e., above 65°S and 65°N) still require validation and should be used with caution, as should those presented for shallow water areas. At a global scale, NPP may also be underestimated in some oceanic regions due to high deep chlorophyll (Chl) maxima (see Table 1 for details).

[1] *https://data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_BGC_001_033/description*
[2] *https://www.st.nmfs.noaa.gov/copepod/*

Table 1: Main sources of product uncertainties, including possible causes and solutions considered for their mitigation (if any).

Source of uncertainty	Cause	Affected area	Solution
Satellite-based NPP is not available	Satellite chlorophyll a data is not available during winter	High latitudes (~65°N) in winter season.	NPP from a biogeo-chemical model is used instead to compensate missing data. A relaxation method is applied to ensure a smooth transition between satellite and model estimates.
Over-estimated NPP	Chl a is likely overestimated in coastal areas due to biases associated to bottom reflectance, sediments and active substances affecting water-leaving radiance (e.g., Blondeau-Patissier et al. 2014)	Shallow waters: coastal areas, internal seas (e.g., Baltic Sea).	Use better estimates of NPP in shallow water (not implemented in this release)
Under-estimated NPP	At a global scale, NPP has been observed to steadily decline since 2011 (i.e., when VIIRS data started to be included), possibly due to an inter-calibration issue between sensors.	Oceanic waters with high deep chlorophyll a maximum.	Currently unknown
Bad estimation of the depths of pelagic layers	The mechanisms that control diel vertical migration need to be explored especially during winter season	High latitudes in winter season	Use a different model of pelagic layer depth (not implemented in this release)
Bad estimation of equatorial circulation in Pacific Ocean	Assimilation of altimetry observations leads to bad estimations of currents and temperature	Western and Central Pacific (especially in the Tropical Warm Pool)	Replace the existing climatology with a climatology from a non-assimilated run. (Not implemented in this release)

Pelagic layer depths:

Since vertical migrations haven been shown to be controlled by light intensity (Benoit-Bird and Wisdoma, 2009), the vertical dimension of the model is simplified and divided into only three layers, whose magnitudes depend on the euphotic zone depth. Available acoustic transects indicate that this simplified 3-layer vertical structure is a correct representation of biological observations (Lehodey et al., 2015) at global scale, or at least for regions with absolute latitudes <65°. At higher latitudes, the mechanisms that control diel vertical migration remain poorly understood, especially during winter season (i.e., in the absence of light).

Zooplankton Biomass:

The climatological database COPEPOD (https://www.st.nmfs.noaa.gov/copepod/) provides standardised mean zooplankton biomass values on a global spatial grid. It contains a compilation of data sampled over a long historical period using a variety of techniques, each with its particular weaknesses and uncertainties that introduce a heterogeneous mix of errors (Moriarty and O'Brien, 2013). The model and its forcing represent another potential source for errors. Nevertheless, as it is the best information available, these observations were used in a first parameter optimisation experiment of the energy transfer coefficient, which was estimated as 0.1668, i.e., close to the commonly used value of 10% (e.g., Ware, 2000).

Micronekton Biomass:

Model outputs are validated using literature estimates of mesopelagic and micronekton biomass. At a global scale, the total biomass of mesopelagic functional groups predicted by the LMTL model is 2.49 Gt. This estimate is in the lower part of the range of estimates found in the literature (0.8 Gt – 16 Gt). At regional levels, the model biomass estimates are consistent with estimates based on net sampling alone or on simultaneous acoustic and net sampling. Conservative correction factors are used to account for trawl efficiency. A conversion coefficient is used to convert carbon to wet weight (WW). The diverse range of oceanic regions from the tropics to temperate waters, and including upwelling and oligotrophic areas, are represented by model values ranging between 0.7 and 15.58 gWW m^{-2} which are comparable to the observational range of 0.3-28.0 gWW m^{-2} (see Table 7).

I.3 Estimated Accuracy Numbers

The following accuracy numbers (Table 2) are estimated for those variables for which in-situ observations were available. For the bottom depths of pelagic layers, observations are extracted from 38kH acoustic transects. For meso-zooplankton density, we use the COPEPOD data base. Comparisons are made using a uniform 5°x5° resolution monthly climatology. For micronekton we compare the total biomass of the mesopelagic functional groups (5 groups) over the global ocean with available estimations.

Table 2: Estimated accuracy numbers of the MICRORYS12V5 variables or diagnostics for which observations are available.

Variable or diagnostics	Metrics	Units	Value or range
Epipelagic layer bottom depth	Mean absolute difference	Meters	**47.3**
Upper mesopelagic layer bottom depth	Mean absolute difference	Meters	**81.8**
Lower mesopelagic layer bottom depth	Mean absolute difference	Meters	**16.1**
Meso-zooplankton biomass density	Mean absolute difference	gC m^{-2}	**0.44**
Mesopelagic micronekton	Absolute difference of the mean total biomass at observation locations	Gt of wet weight	**[0.13;16.3]**

II PRODUCTION SYSTEM DESCRIPTION

The product GLOBAL_MULTIYEAR_BGC_001_033 described in this document has been produced at CLS (Toulouse, France). It is a global simulation using the Low and Mid-Trophic module of the Spatial Ecosystem and Population Dynamic Model (SEAPODYM-LMTL - http://www.seapodym.eu.) It is forced by daily fields of ocean and satellite derived NPP. Ocean forcings come from the global ocean eddy-resolving 1/12° horizontal resolution and 50 vertical levels reanalysis covering the altimetry era 1993-2023 (GLORYS12V1 which corresponds to the Copernicus Marine Service product GLOBAL_REANALYSIS_PHY_001_030). NPP is computed from chlorophyl *a* concentrations provided by the Copernicus Marine Service OCEANCOLOUR_GLO_BGC_L4_MY_009_104 product. There is no data assimilation in this product.

II.1 SEAPODYM-LMTL

II.1.1 Short description

SEAPODYM is an ecosystem model that simulates the change in biomass over time and space of several lower (meso-zooplankton) and mid trophic (micronekton) functional groups and target fish species, resolved by age class ranging from larvae to oldest adults (Lehodey et al., 2008). Traditionally, micronekton has been defined as organisms in the 2cm-20cm size range and contains a large diversity of fish, crustaceans, squid, and gelatinous species (all taxa are grouped into different functional groups based on their diel migratory behaviour). Note that micronekton also includes small (young) individuals of species with large growth potential. SEAPODYM is based on a system of advection-diffusion-reaction equations that are forced by ocean currents, temperature, net primary productivity, and euphotic depth. Horizontal ocean currents control the transport of organisms, while temperature controls organism development and mortality. NPP serves as the biomass source. Euphotic depth controls the position (and thickness) of the three pelagic layers, chosen as a simplified representation of the water column based on literature recommendations (Lehodey et al., 2015).

The GLOBAL_MULTIYEAR_BGC_001_033 product uses the LMTL module of SEAPODYM in the following configuration:

- Six independent functional groups of micronekton inhabiting three broad pelagic zones: the epipelagic layer, upper mesopelagic layer, and lower mesopelagic layer.
- A single functional group of meso-zooplankton inhabits the epipelagic layer.

Micronekton perform diel vertical migrations (DVM). Given the structural importance of this migration, the functional groups are defined based on the daily vertical migration behaviour of organisms: Three groups contain organisms that remain in the same layer throughout the day (24h) (called non-migrant groups below) while the three remaining groups contain species that migrate between layers over the course of a day-night cycle (called migrant groups) (Lehodey et al., 2010; 2015). summarizes the names and nomenclature used for micronekton.

Table 3: Names and nomenclature used for the six functional groups of micronekton. Day/Night layer refers to the vertical position of corresponding functional groups during day/night time, with 1 being the surface layer and 3 the bottom layer.

Functional group name	Day layer	Night layer	Notations used
Epipelagic micronekton	1	1	D1N1 or 1.1
Upper mesopelagic micronekton	2	2	D2N2 or 2.2
Migrant upper mesopelagic micronekton	2	1	D2N1 or 2.1
Lower mesopelagic micronekton	3	3	D3N3 or 3.3
Migrant lower mesopelagic micronekton	3	2	D3N2 or 3.2
Highly migrant lower mesopelagic micronekton	3	1	D3N1 or 3.1

Acoustic observations and net samplings suggest that the pelagic zones (also called bio-acoustic layers) are distributed between the surface and ~ 800 meters depth. Euphotic depth was selected as an explanatory variable due to the strong evidence that vertical migrations are controlled by light intensity, at least in the tropical and temperate regions. The euphotic depth is used to define the vertical boundaries of the three pelagic layers as follows (Lehodey et al., 2015):

- Bottom depth of epipelagic layer = 1.5 x euphotic depth,
- Bottom depth of upper mesopelagic layer = 4.5 x euphotic depth,
- Bottom depth of lower mesopelagic layer = MIN (10.5 x euphotic depth, 1000m),

where euphotic depth is given in meters (cf., Figure 1).

Meso-zooplankton is modelled as a single group inhabiting the epipelagic layer (Conchon, 2015).

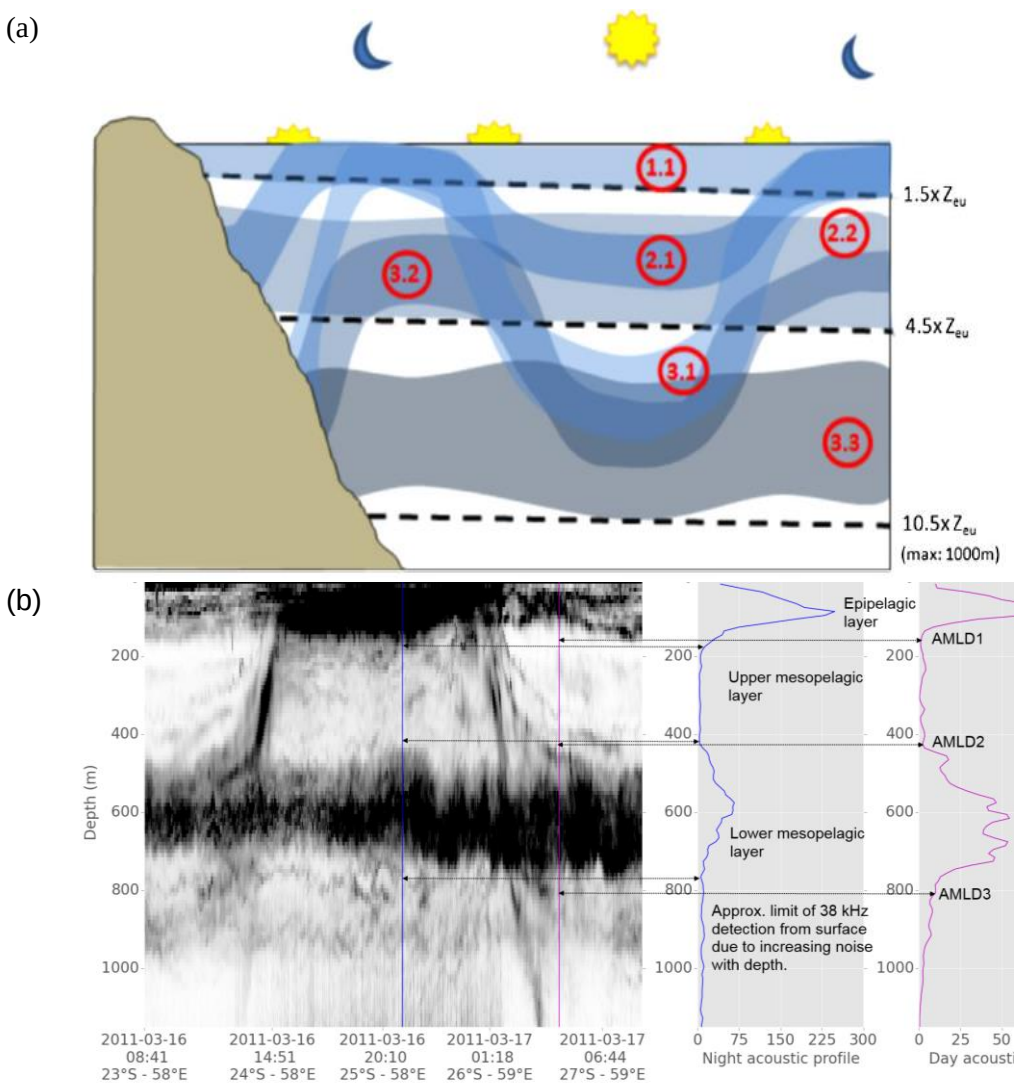

Figure 1: (a): Identification of MTL functional groups based on their diel vertical migrations seen in the acoustic echogram (bottom left) and conceptual model (bottom right) (from Lehodey et al., 2015). Grey shaded areas illustrate the position of non-migrant functional groups while blue shaded areas illustrate position of migrant functional groups. See Table 3 for functional group notation (in red circles) used. (b): Example of an acoustic transect showing bio-acoustic layers and detected layer depths (from Conchon, 2015).

II.1.2 Specific model configuration and tunings

The GLOBAL_MULTIYEAR_BGC_001_033 simulation using the SEAPODYM-LMTL ecosystem model is based on version 4.0.0. It is forced offline by daily fields from the ocean dynamical assimilated simulation GLORYS12V1. There is no data assimilation, and the parameters are tuned (Table 4) based on available information (cf. Section III). There are nine parameters to control the biological processes: six coefficients of energy transfer from primary production to mid-trophic level functional groups, two parameters for the relationship between time of development (mortality, recruitment) and water temperature, and one diffusion coefficient. For meso-zooplankton, there is only one energy transfer efficiency coefficient and relationships between time of development and water temperature for mortality and recruitment. The same diffusion coefficient is used for all groups. The parameterisation of

time development-temperature (Figure 2) is based on results published in the literature (Huntley and Lopez, 1992; Gillooly, 2002; Lehodey et al., 2010). Therefore, the main uncertainty remains associated with the energy transfer efficiency coefficients that control the total biomass. Since micronekton biomass is expressed in wet weight, the different factors proposed in the literature for converting from carbon to wet weight may also lead to substantial differences. Biomass of meso-zooplankton is expressed in grams of carbon with the conversion factor to wet weight provided for user information. Furthermore, the model is highly sensitive to the forcing fields, especially NPP. For meso-zooplankton, a maximum likelihood approach was used to obtain a parameterisation of the energy transfer efficiency coefficient using the observational data described in Section III, averaged over a monthly 5° x 5° resolution (resulting in a total of 5023 observations). Model meso-zooplankton biomass estimates (gC m^{-2}) were divided by the epipelagic layer depth and converted to mg for comparison. The transfer efficiency was estimated as 0.1668, close to the commonly used value of 10% (e.g., Ware, 2000).

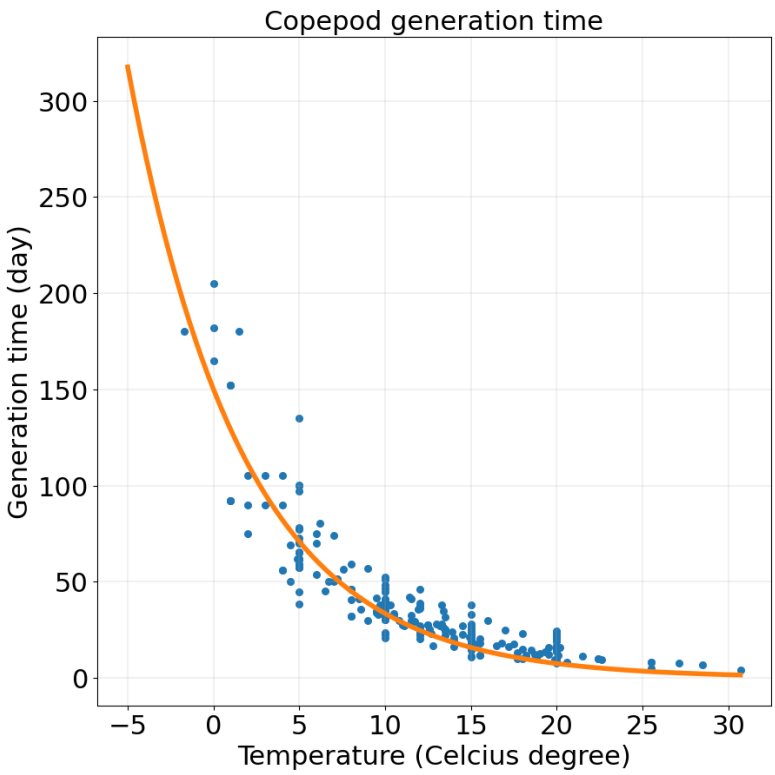

Figure 2 Published values of copepod generation time as a function of temperature (blue points) (data compiled by Huntley and Lopez, 1992; Conchon, 2015) with the analytical relationship implemented in MICRORYS12V5 (solid line).

Table 4 : Parameterisations used in the GLOBAL_REANALYSIS_BIO_001_033 product.

	Description	Low Trophic Level	Mid Trophic Level	Units
tr_{max}	Recruitment time at 0°C	10.38	527	d
tr_{curv}	Coefficient of the exponential decreasing relationship for recruitment time	-0.11	-0.125	-
$\left(\frac{1}{\lambda}\right)_{max}$	Age at maturity (equivalent to 1/λ) at 0°C	150	2109	d
$\left(\frac{1}{\lambda}\right)_{curv}$	Coefficient of the exponential decreasing relationship for 1/λ	-0.15	-0.125	-
E	Total energy transfer efficiency from NPP to trophic levels	0.1668	0.0042	-
cww	Conversion factor (g carbon to g wet weight)	10.0	11.9	-
$E'_{1.1}$	Energy transfer efficiency to functional group 1.1	1	0.17	-
$E'_{2.1}$	Energy transfer efficiency to functional group 2.1	-	0.22	-
$E'_{2.2}$	Energy transfer efficiency to functional group 2.2	-	0.10	-
$E'_{3.1}$	Energy transfer efficiency to functional group 3.1	-	0.20	-
$E'_{3.2}$	Energy transfer efficiency to functional group 3.2	-	0.13	-
$E'_{3.3}$	Energy transfer efficiency to functional group 3.3	-	0.18	-
D	Diffusion (both water and random movements of organisms)	5	5	$m^2 \cdot s^{-1}$

SEAPODYM uses a 1/12° resolution Yin-Yang overset computation grid with 2° overlap (Kageyama *et. al. 2004) (see* Figure 3*).* The vertical dimension is simplified using a total of 3 layers (epi- and upper/lower mesopelagic) between the surface and ~800 m depth (Figure 1). The water temperature and horizontal (U and V) current velocities are averaged within each layer. The vertical boundary layers are defined dynamically and proportionally to the euphotic depth.

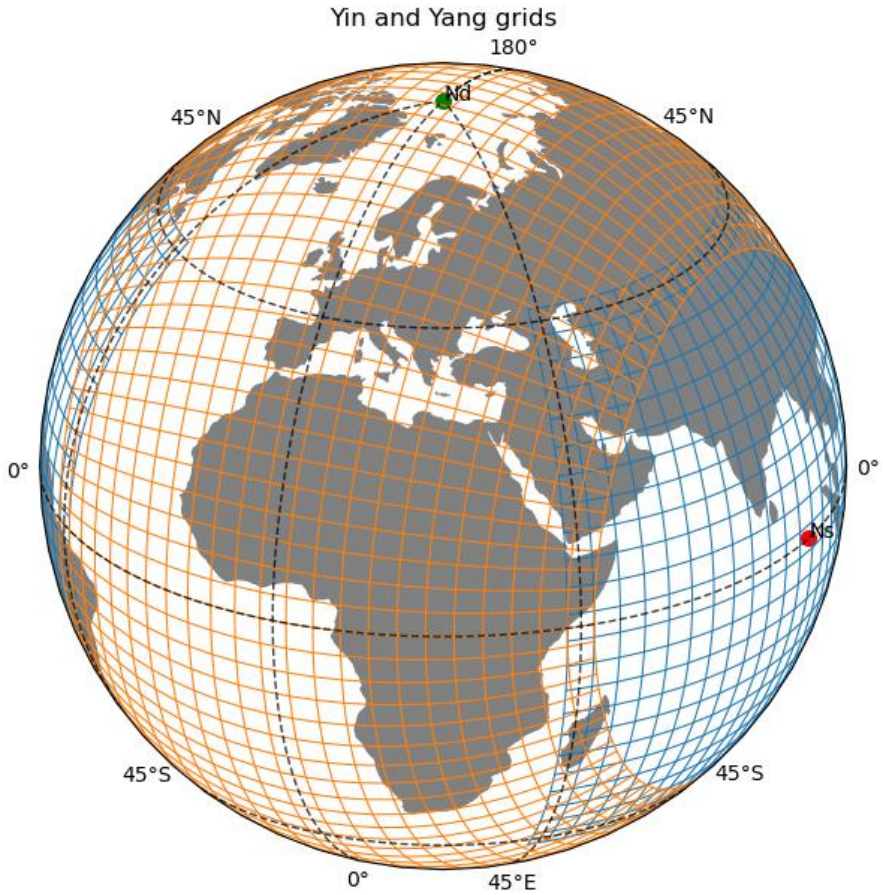

Figure 3. Example of Yin (blue) and Yang (orange) grids with 2° overlap. The Yin grid is a regular latitude/longitude grid covering the domain [-47°N, 47°N] latitudes and [43°E, 317°E] longitudes. The Yang grid is the corresponding rotated pole grid with North pole located at longitude 90° and latitude 0° in the geographical latitude/longitude coordinate system (red dot).

The GLOBAL_MULTIYEAR_BGC_001_033 simulation is initialized in January 1998, which corresponds to the beginning of global satellite observations with SeaWiFS, providing ocean colour data which serves as a proxy to estimate primary production.

Initial conditions come from a one-year spinup for zooplankton and from a ten-years spinup for micronekton. Spinups are based on a climatological series.

II.1.3 Limitations

SEAPODYM has been developed for ocean pelagic ecosystems and does not include any representation of the benthos. In shallow waters with only one or two vertical layers, the energy flux from the missing functional groups is redistributed equally between the remaining groups, leading to potential biomass overestimations in these areas.

The simplified vertical structure of the model does not account for vertical velocities that can produce local convergences or divergences of organisms, especially at the scale of eddies or abrupt topography (e.g., seamounts). These dynamics may influence the distribution and behaviour of organisms, but they are not captured in the current model configuration.

II.2 Ocean physics forcings: temperature and currents

Daily 3D temperature and horizontal velocity fields are used to force the SEAPODYM-LMTL model. These come from the GLORYS12V1 simulation which is the Copernicus Marine Service global ocean eddy-resolving (1/12° horizontal resolution, 50 vertical levels) reanalysis covering the altimetry period (1993 onward) (GLOBAL_MULTIYEAR_PHY_001_030 product). It is based largely on the current Copernicus Marine Service real-time global forecasting system. The model component is the NEMO platform driven at the surface by ECMWF ERA-Interim before 2018 and from 2018 onward by ERA5 reanalyses. Observations are assimilated using a reduced-order Kalman filter. Along-track altimetry data (sea level anomaly), sea surface temperature, and sea ice concentration, as well as in situ temperature and salinity (vertical profiles) are jointly assimilated. Moreover, a 3D-VAR scheme provides a correction for the slowly evolving large-scale biases in both temperature and salinity.

II.3 Ocean biogeochemical forcings: NPP and euphotic depth

NPP and the associated euphotic depth are computed using the Vertically Generalized Production Model (VGPM) of Behrenfeld and Falkowski (1997) which is based on the Satellite Observations reprocessed Global Ocean Chlorophyll product COPERNICUS-GLOBCOLOUR OCEANCOLOUR_GLO_BGC_L4_MY_009_104, NOAA Optimum Interpolation SST V2.1 AVHRR only (https://www.ncei.noaa.gov/products/optimum-interpolation-sst) and photosynthetically active radiation (PAR) (from ERA-INTERIM reanalysis (ECMWF) for 1998-2018 and from NCEP from 2019 to 2022) . During winter seasons, there is no reliable satellite chlorophyll data for latitudes above ~45° due to sun glint, cloud coverage, and low light levels. Hence, the datasets have been extended (relaxed) to primary production predicted by the biogeochemical product GLOBAL_REANALYSIS_BIO_001_029. Since the data gaps occur during winter when primary production is extremely low in these regions, the contribution of this biogeochemical input is extremely low. The GLOBAL_MULTIYEAR_BGC_001_033 simulation product is provided for latitudes between 80°S-90°N but still requires validation at high latitudes (>65°).

III VALIDATION FRAMEWORK

The products assessed are vertical pelagic layer boundaries and meso-zooplankton and micronekton biomass densities.

Pelagic layer boundaries are compared to those extracted from 38 kHz acoustic data. Datasets mainly come from Integrated Marine Observing System (IMOS) and British Antarctic Survey (BAS) transects (Figure 4). A K-means unsupervised classification algorithm (machine learning) was used to estimate the bottom depth of each pelagic layer using a total of 194350 observations. Estimated depths have been validated (and eventually manually corrected) by an expert.

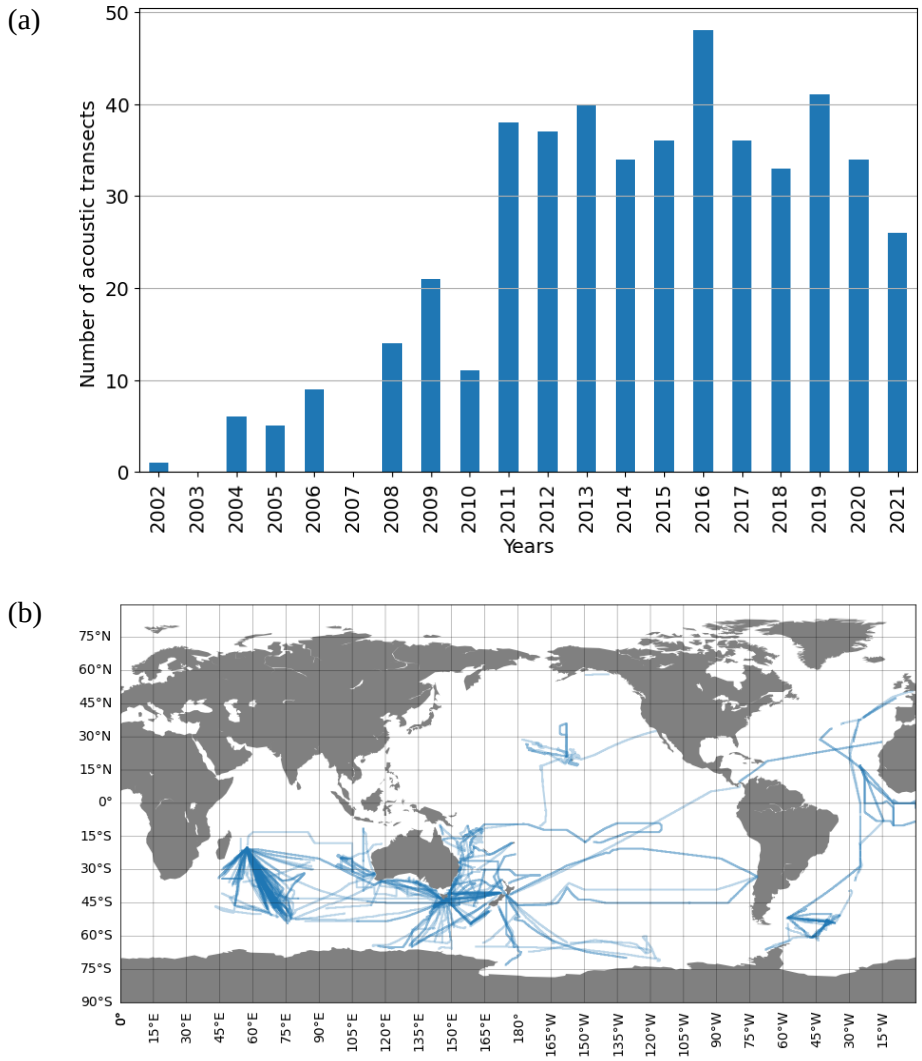

Figure 4: Acoustic transect data sources used in the validation of model estimates of pelagic layer depths. (a): number of transects per year for each data source. (b): map showing transect locations.

Meso-zooplankton biomass product is compared to the monthly climatological series from the COPEPOD database which provides over 150,000 standardised observations at a global scale (Moriarty and O'Brien, 2013) (Figure 5). These data are compiled from long historical time series that used different sampling approaches and units to record their results (e.g., wet mass, dry mass, volume). These needed to be standardised to a common unit (gC m^{-2}) and integrated over the depth interval (0–200 m), while converting them to a common mesh size (samples collected with 150–200 µm mesh sizes were converted to 330 µm). Despite the considerable effort that went into the standardisation and collation of this dataset, the lack of consistency in measurement approaches over time represents a source of uncertainty. In an attempt to reduce uncertainty, we applied a coarser resolution of 5°×5° and excluded data collected in shallow waters where the total water depth was <100 m, yielding a total of 5023 observations.

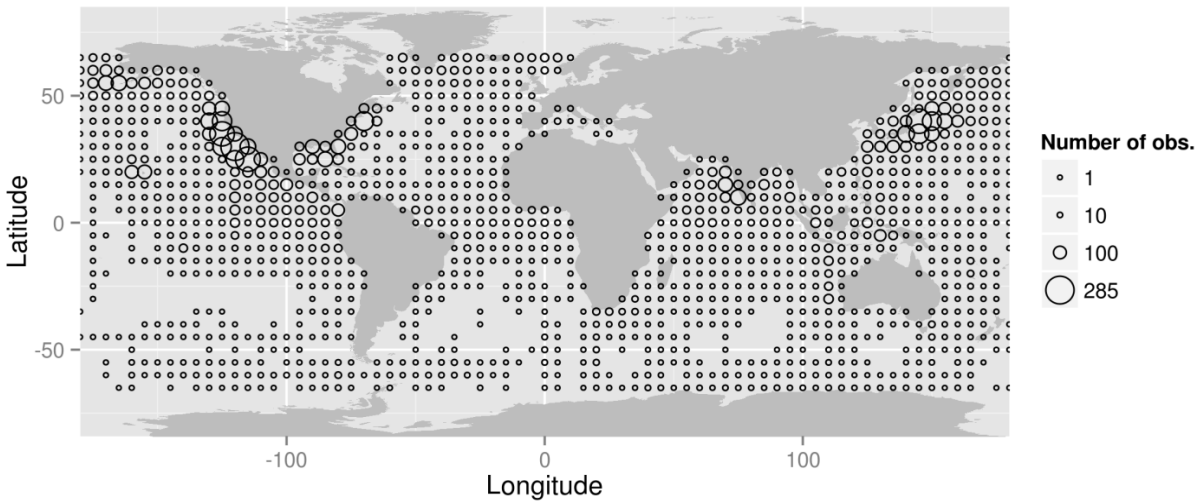

Figure 5: Location of meso-zooplankton biomass observations in the COPEPOD monthly climatology.

Observational data on micronekton remains scarce, unfortunately. Traditionally, micronekton biomass has been estimated using net trawling. However, corrections must be made to account for sampling efficiency. Combined acoustic-trawl surveys use trawl sampling to validate acoustic measurements. Target strengths that depend on species and length are used in an acoustic observation model to convert the measured signals into biomass. This method is expensive and usually deployed for economically important species. The few existing studies that targeted total mesopelagic biomass have been compiled and are used here for the model validation. Predictions from other ecosystem models are also of interest as they provide global estimates based on independent methods. Finally, acoustic data only is still useful for validation, to assess relative variability over time in frequently sampled areas, and to validate the depths of biological layers.

IV VALIDATION RESULTS

IV.1 Pelagic layer boundaries

IV.1.1 Relevant diagnostics, based on metrics

A K-means unsupervised classification algorithm (machine learning) was used to estimate the bottom depth of each pelagic layer, using 193,682 observations. Observations were validated (and when necessary manually corrected) by an expert. We compare the (absolute differences in) bottom depth of each layer with those obtained based on the euphotic depth-based expression in Section II.1.1 (Figure 6).

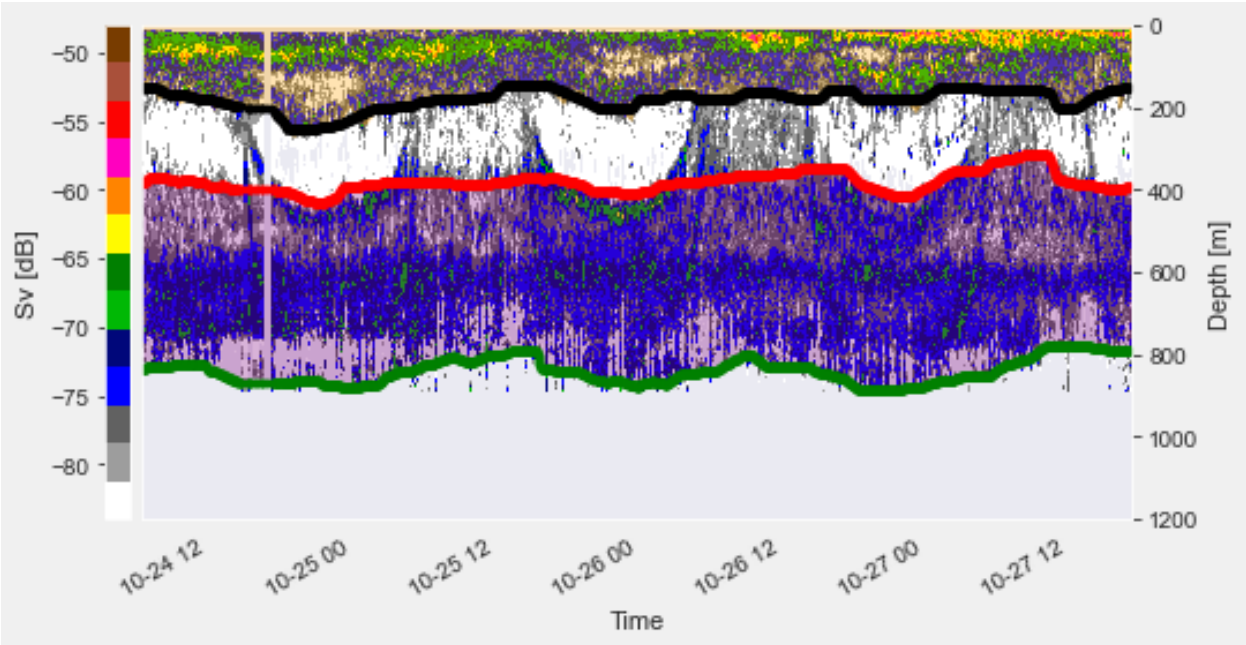

Figure 6: Visualisation of a 38kH acoustic transect with the pelagic layer partitioning obtained using a K-means algorithm. The horizontal lines mark the bottom depths of the epipelagic layer (black), upper-mesopelagic layer (red), and lower-mesopelagic layer (green).

IV.1.2 Synthesis of product quality for this variable

The simplified 3-layer vertical structure yields a correct representation of biological observations at a global scale, specifically for regions with absolute latitudes <65°. Table 5 and Figure 7 summarize the results of the statistical analysis of absolute differences between observed and modelled pelagic bottom depths for each layer.

Table 5: Statistics of absolute differences between observations and model estimates of pelagic layer bottom depths (observations – model estimates) in metres.

Layer	Minimum	Standard deviation	Mean	Maximum
Epipelagic	-91.30	54.20	**47.26**	280
Upper mesopelagic	-156.34	69.38	**81.85**	383.67
Lower mesopelagic	-505.82	175.62	**16.11**	711.82

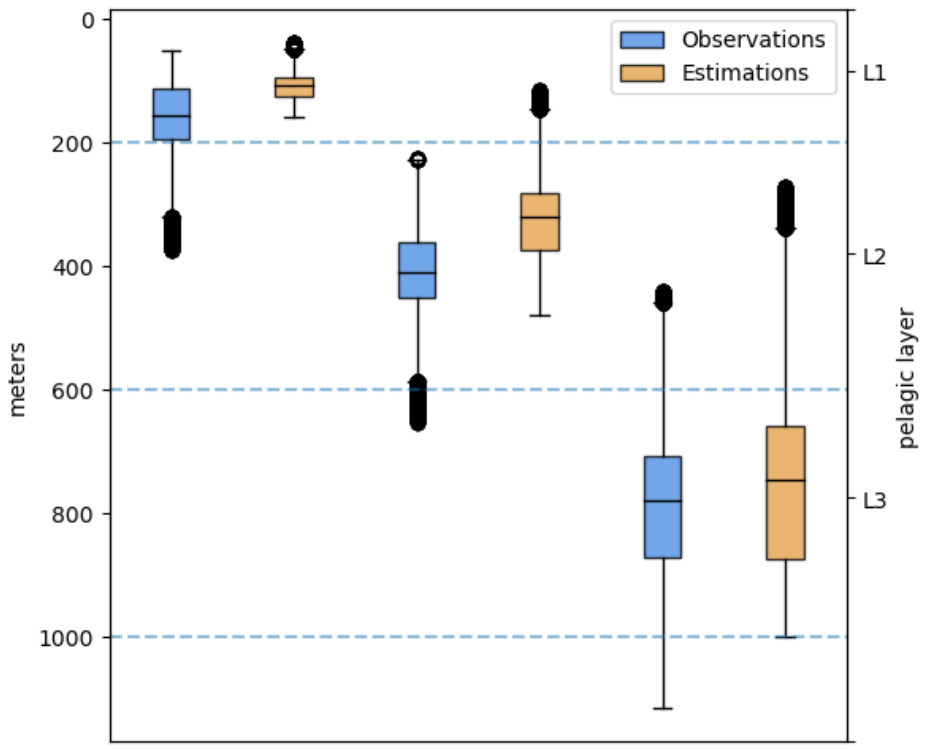

Figure 7: Boxplot of observed (blue) and estimated (orange) pelagic layer bottom depths. Dashed lines represent the pelagic layer bottom depths as defined by Kaiser et. al. (2011).

IV.2 Meso-zooplankton

IV.2.1 Relevant diagnostics, based on metrics.

Current knowledge and quantitative information on meso-zooplankton global distribution and biomass remains limited. The COPEPOD database is the only global dataset providing a compilation of historical sampling efforts measuring meso-zooplankton biomass (Moriarty & O'Brien, 2013). Despite the effort devoted to the standardisation of these data, there are numerous and diverse sources of errors. Of the total of 153,163 biomass values, approx. 2% were originally recorded as dry mass, approx. 26% as wet mass, approx. 5% as settled volume, and approximately 68% as displacement volume. These have been converted from their original units to carbon biomass using a variety of non-linear biomass conversions available in the literature. They also had to be "mesh-adjusted" to a single mesh size of meso-

zooplankton net. Finally, depth-integrated values were simply extrapolated, thereby ignoring potentially large vertical variability in biomass distribution. Distributions of observed and estimated biomass densities range from 0 to 6 gC m^{-2} and show a log-normal shape (Figure 8b). The mean biomass density is lower for observations (0.76 gC m^{-2}) than for model estimations (1.19 gC m^{-2}) with lower standard deviations (0.86 vs 1.27, respectively) (Figure 8a). A bi-modal distribution is visible for MICRORYS12V5 estimations with a small bump between 0.8 and 4 gC m^{-2} (Figure 8b).

The spatial biomass distribution seems relatively consistent throughout the year. Largest absolute values of the bias are visible at high latitudes or in areas with narrow straits of sea surrounded by land. Those areas are particularly difficult to model. Slightly negative biases are visible at the equator while slightly positive values are visible in the subtropics (Figure 14[3]).

The climatology may be biased in some regions if the observations have been collected during a year with abnormal conditions (e.g., El Niño southern oscillation). NPP may also be a source of errors. For our validation, we removed the 99th percentile from the observational data (this represents a removal of 180 observations from a total of 5023). Validation was performed using 4843 observations, yielding a relatively poor linear fit between observations and model outputs (Figure 14 and Figure 15; R^2=0.11).

The absolute difference (observation minus estimation) ranges from -5.33 to 5.63 gC m^{-2} and does not show any systematic bias (median of 0.083 gC m^{-2} ; Figure 16).

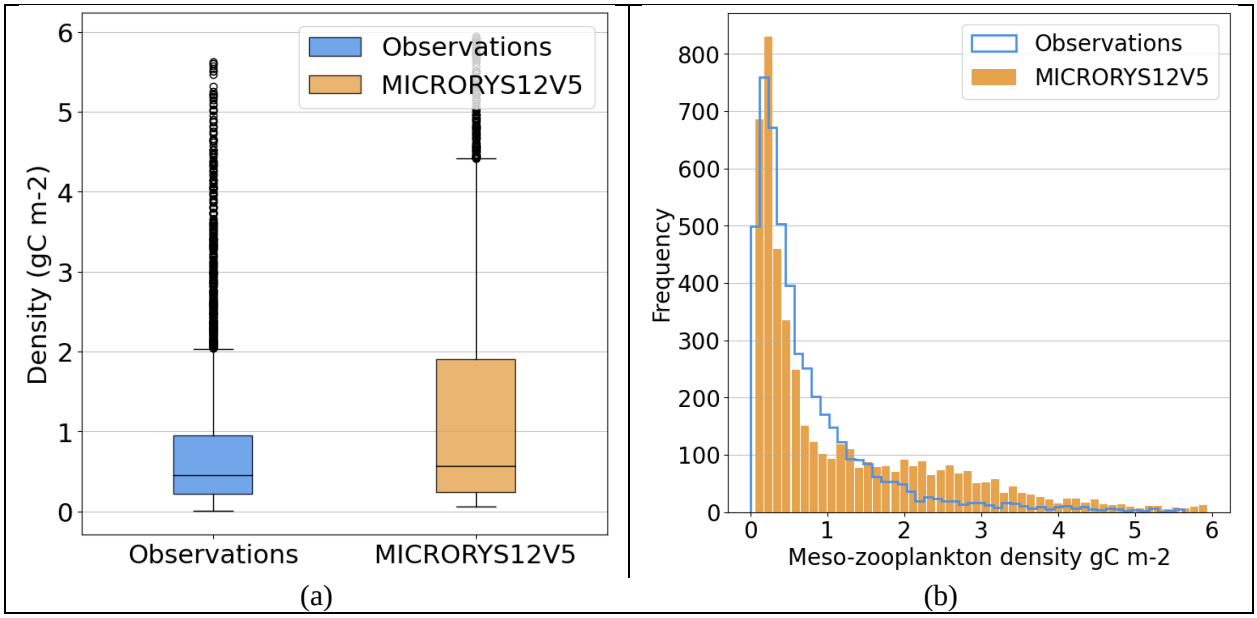

Figure 8:Distribution of 5°x5° monthly climatology of observed and modelled meso-zooplankton density. Blue: data from COPEPOD database. Orange: MICRORYS12V5 estimation.

[33] Note that Figure 9 has 12 panels on six pages.

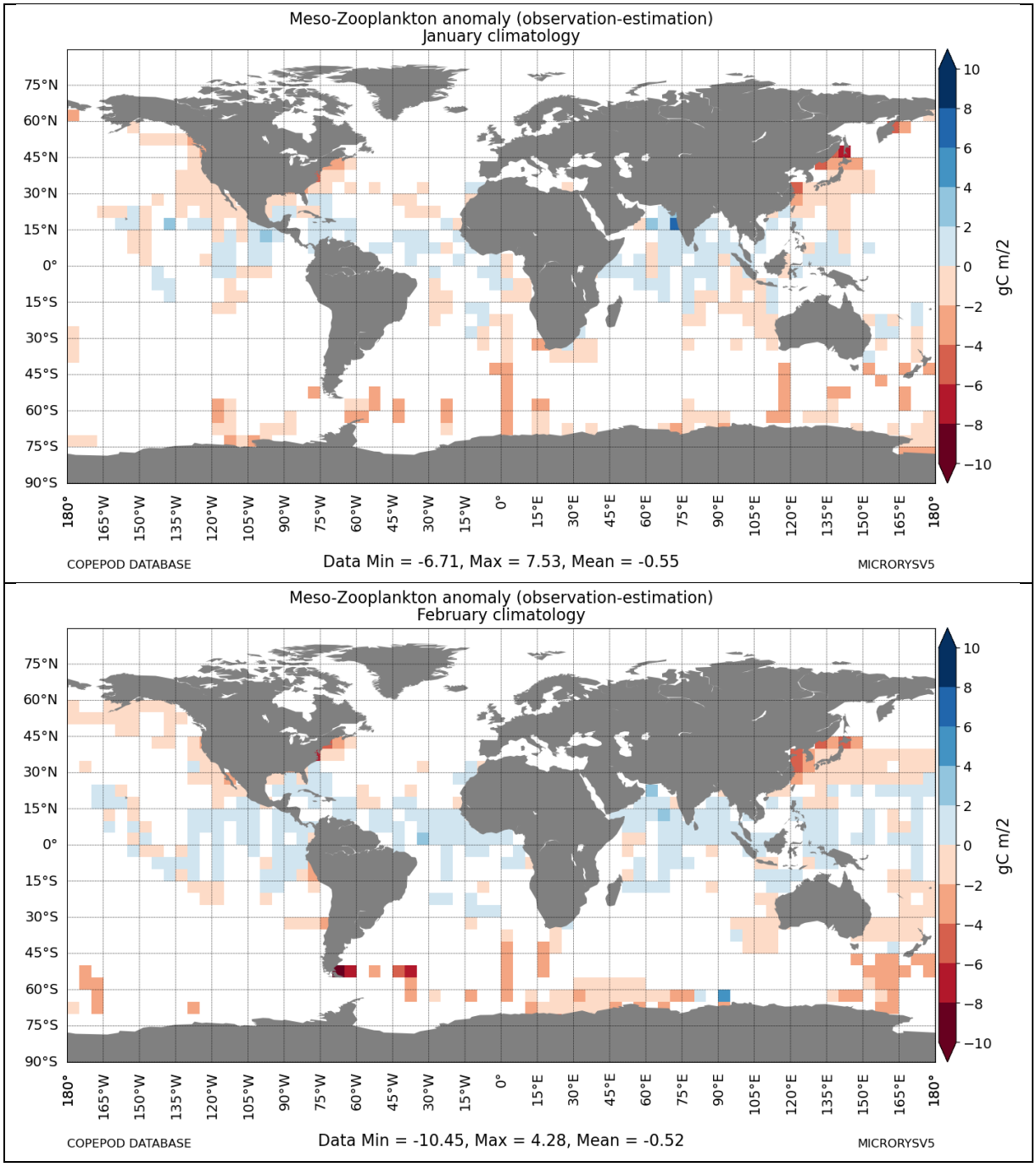

Figure 9 :Maps of absolute differences between 5x5° monthly (January to December) climatology of observed and MICRORYS12V5 meso-zooplankton biomass density at locations for which observations are available. Observations come from the COPEPOD database. From top to bottom: January to December climatology. Each colour bar has the same range [-10,10] gC m^{-2} to facilitate comparisons.

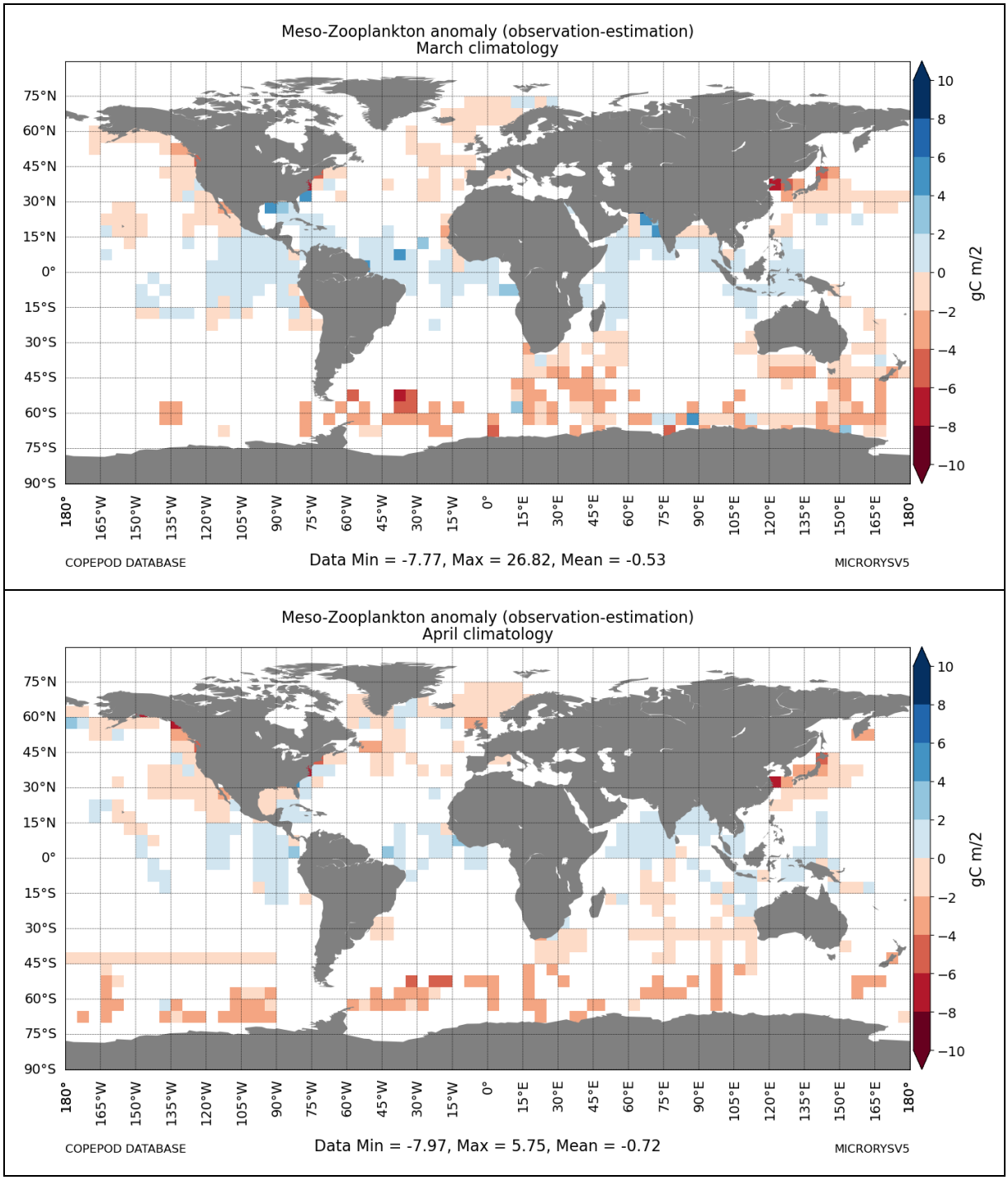

Figure 10 cont: Maps of absolute differences between 5x5° monthly (January to December) climatology of observed and MICRORYS12V5 meso-zooplankton biomass density at locations for which observations are available. Observations come from the COPEPOD database. From top to bottom: January to December climatology. Each colour bar has the same range [-10,10] gC m^{-2} to facilitate comparisons.

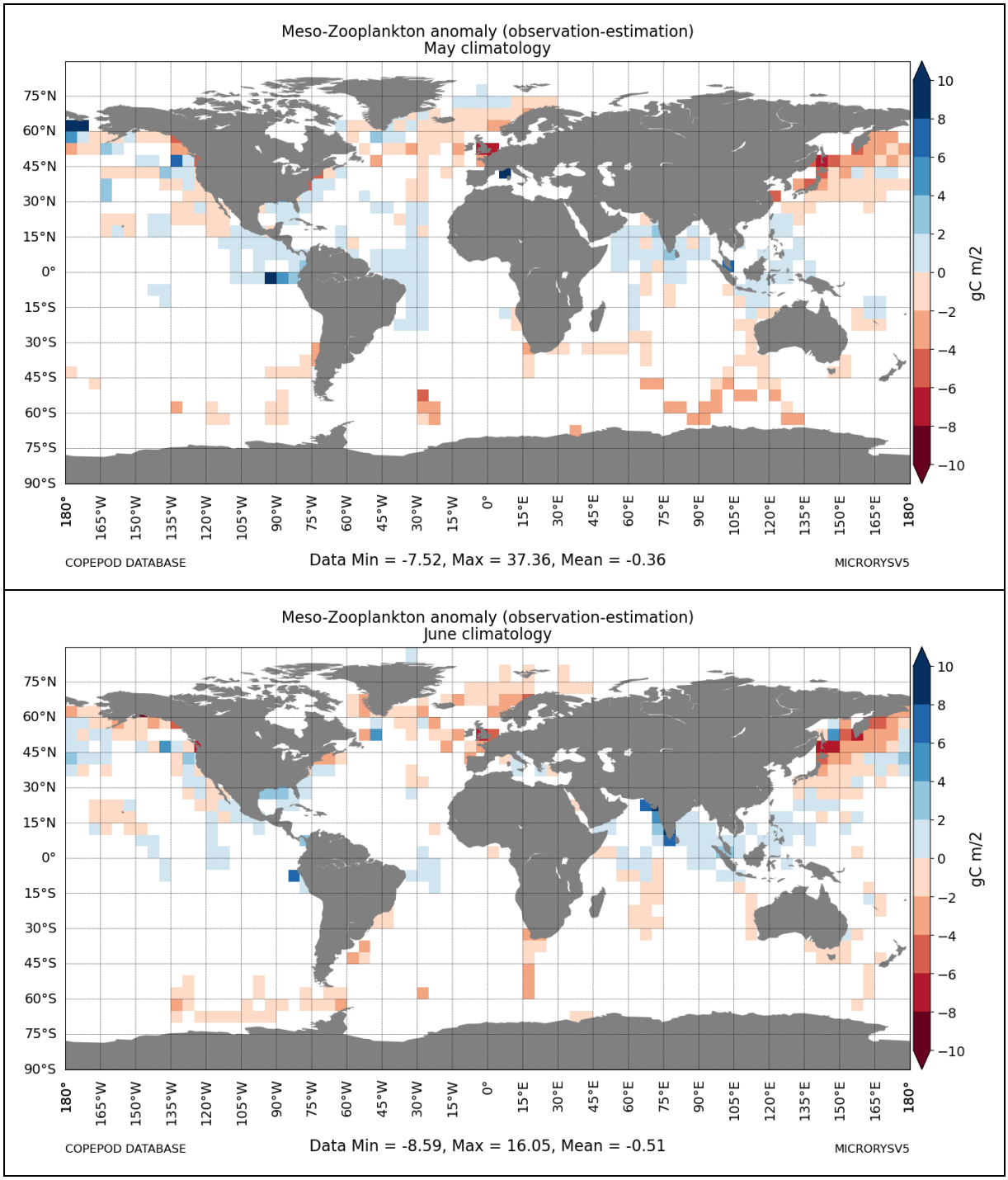

Figure 11 cont: Maps of absolute differences between 5x5° monthly (January to December) climatology of observed and MICRORYS12V5 meso-zooplankton biomass density at locations for which observations are available. Observations come from the COPEPOD database. From top to bottom: January to December climatology. Each colour bar has the same range [-10,10] gC m^{-2} to facilitate comparisons.

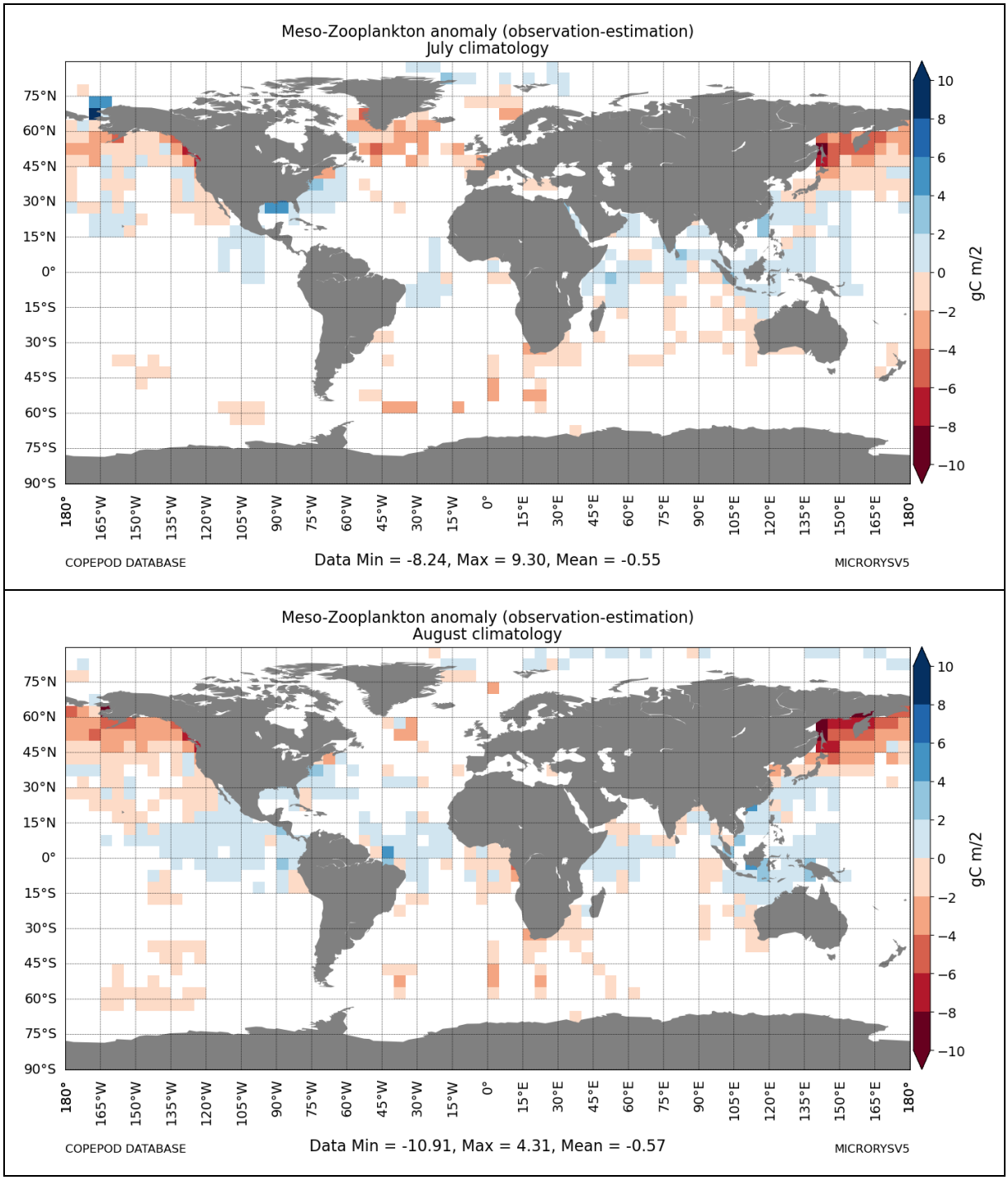

Figure 12 cont: Maps of absolute differences between 5x5° monthly (January to December) climatology of observed and MICRORYS12V5 meso-zooplankton biomass density at locations for which observations are available. Observations come from the COPEPOD database. From top to bottom: January to December climatology. Each colour bar has the same range [-10,10] gC m^{-2} to facilitate comparisons.

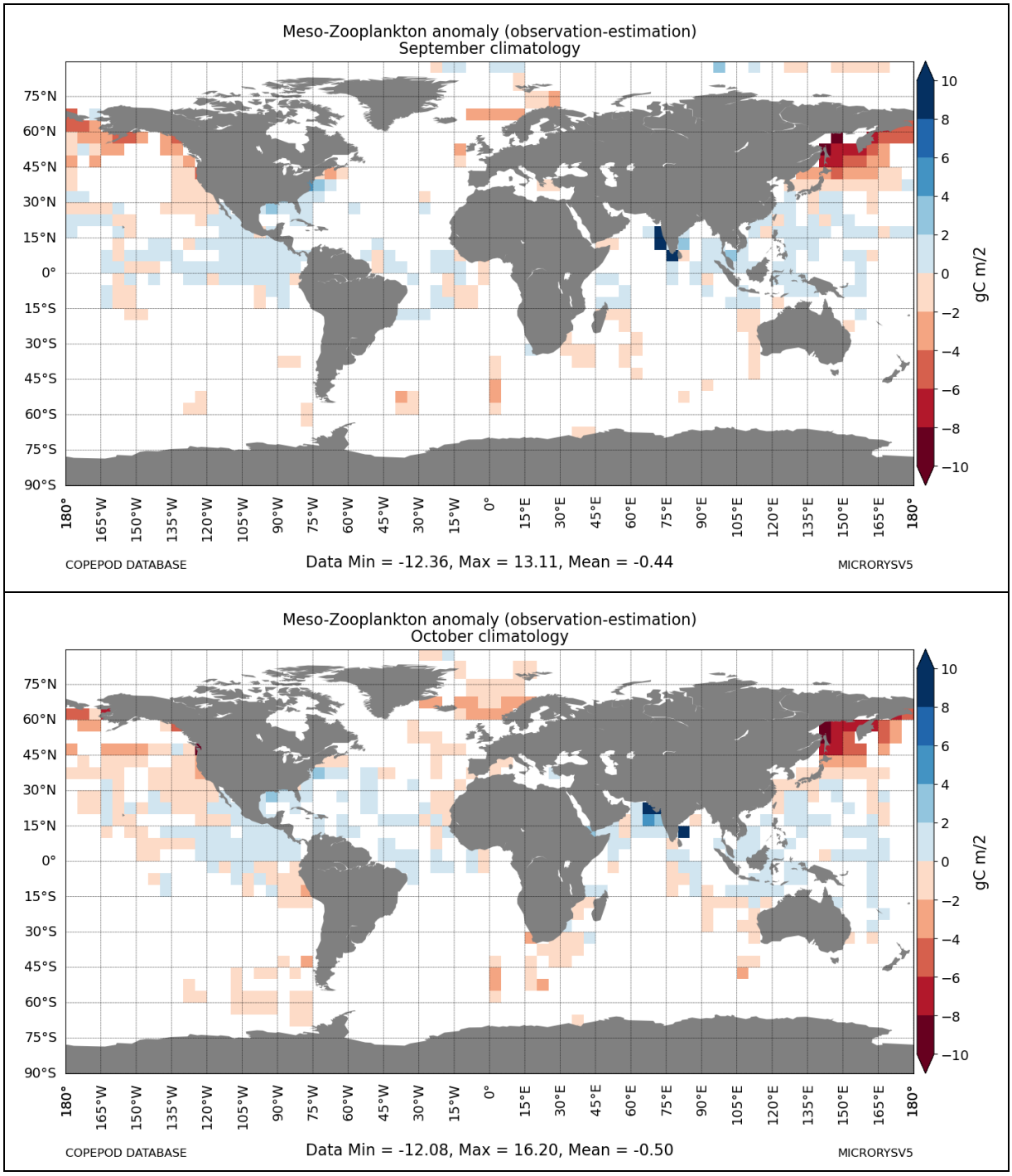

Figure 13 cont: Maps of absolute differences between 5x5° monthly (January to December) climatology of observed and MICRORYS12V5 meso-zooplankton biomass density at locations for which observations are available. Observations come from the COPEPOD database. From top to bottom: January to December climatology. Each colour bar has the same range [-10,10] gC m^{-2} to facilitate comparisons.

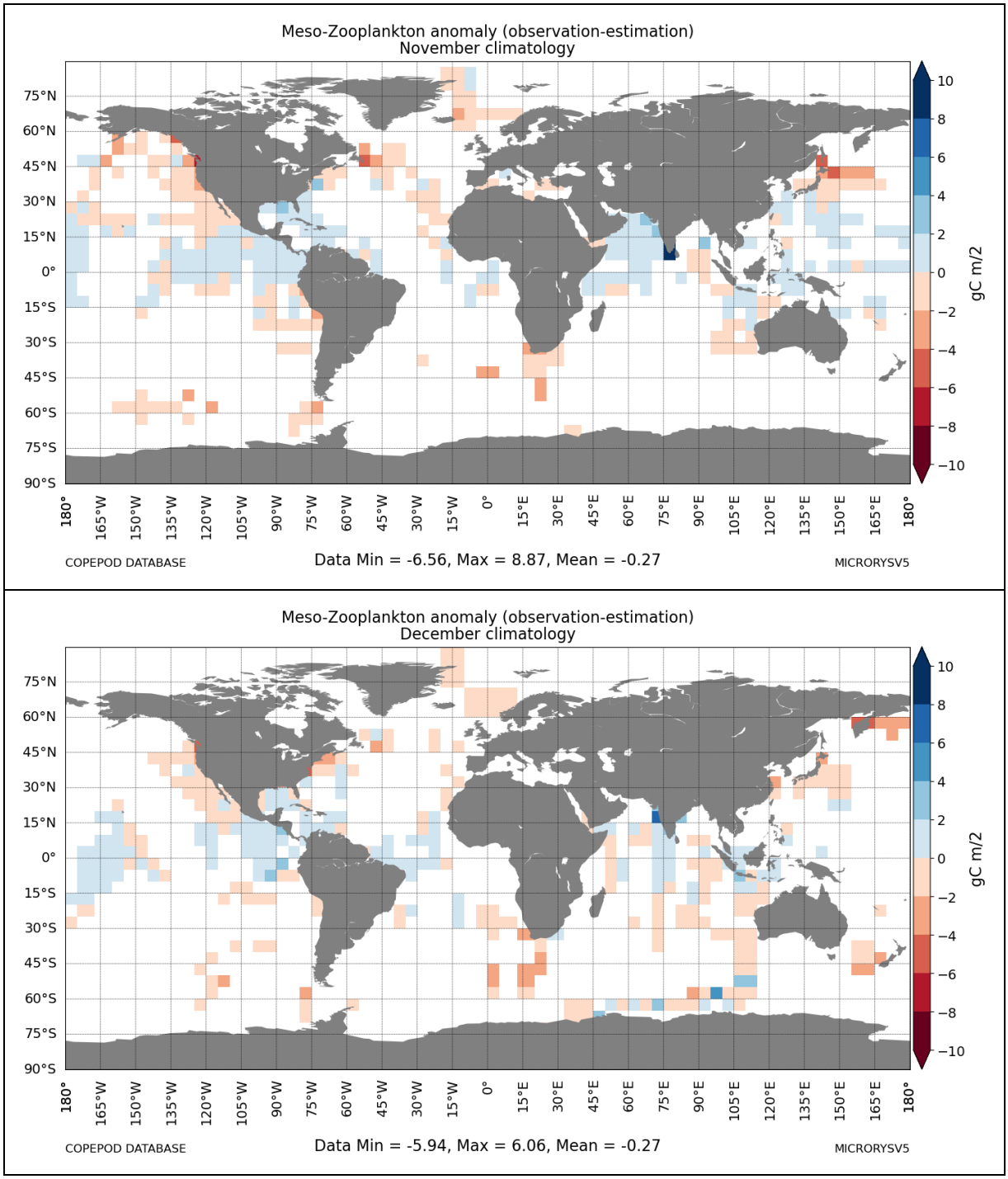

Figure 14 cont: Maps of absolute differences between 5x5° monthly (January to December) climatology of observed and MICRORYS12V5 meso-zooplankton biomass density at locations for which observations are available. Observations come from the COPEPOD database. From top to bottom: January to December climatology. Each colour bar has the same range [-10,10] gC m^{-2} to facilitate comparisons.

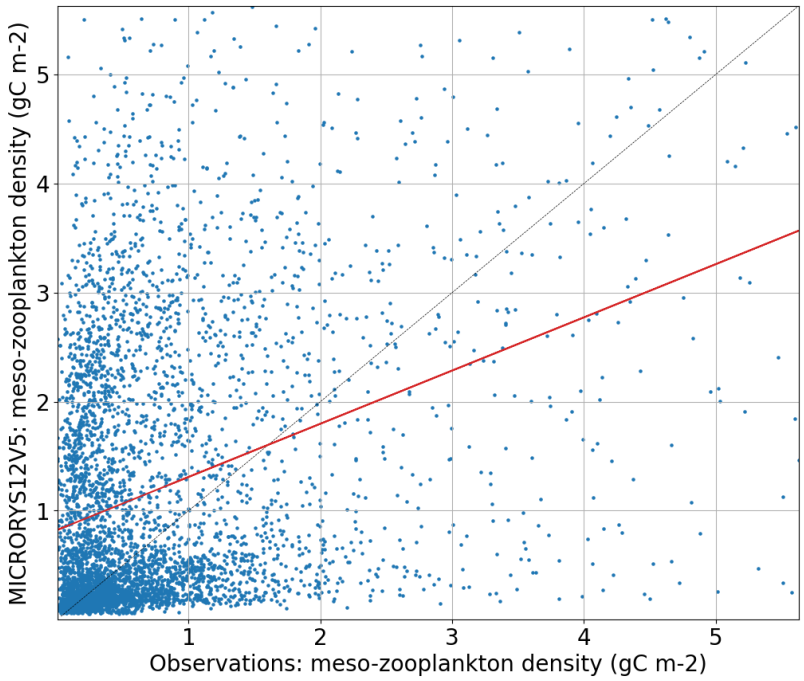

Figure 15: Observed vs estimated meso-zooplankton biomass monthly climatological densities at 5°x5° spatial resolution. Red line: linear regression between observation and estimation. Dashed black line: identity function.

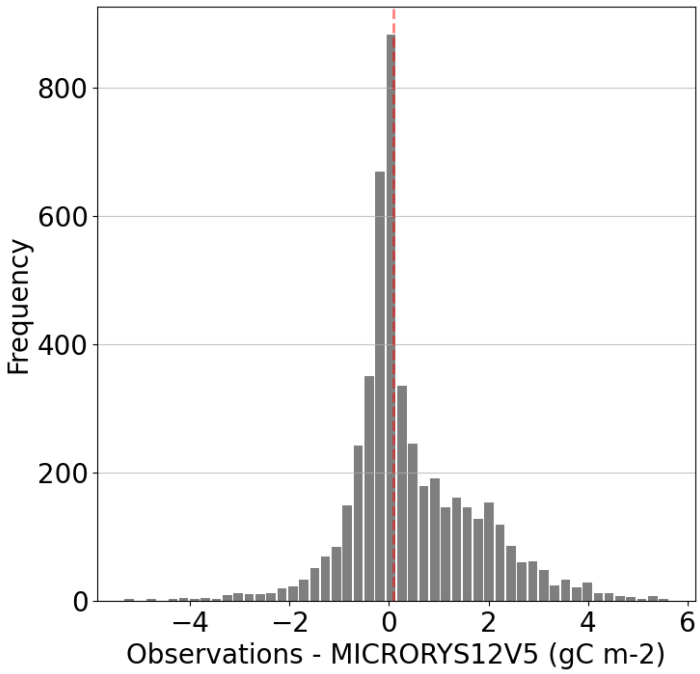

Figure 16: Distribution of absolute differences (observation minus estimation) of monthly climatological data at 5°x5° resolution. Red dasedh line shows the median.

IV.2.2 Synthesis of product quality for meso-zooplankton.

The model meso-zooplankton biomass climatological distributions have been compared to the only available global database (COPEPOD). The comparison is somewhat fraught by the many uncertainties and wide variety of potential sources of error (e.g., model forcing, historical observations using various units and approaches that needed to be standardised, etc.). As a result, the simulation results should be regarded as a first reasonable representation of global meso-zooplankton biomass distributions serving as a foundation for future improvements.

IV.3 Micronekton

IV.3.1 Relevant diagnostics, based on metrics.

Model outputs are validated using literature values of mesopelagic micronekton biomass. At a global scale, the biomass of *mesopelagic* functional groups has been compared to some existing estimates (Table 6). The MTL model predicted an annual mean total biomass for the mesopelagic functional groups in the range [2.33Gt;2.58Gt] with a mean of 2.49 Gt over the considered period (1998-2023). This estimate is in the lower part of the range of known estimates in the literature (0.8 Gt – 16 Gt).

Table 6: Estimates of global (open ocean) mesopelagic fish biomass

Estimate	Reference	Method
0.8 Gt	Gjøsæter and Kawaguchi (1980)	Ocean trawl data
11 to 15 Gt (40°N-40°S)	Irigoien *et al.* (2014)	Acoustics
< 1.4 Gt	Jennings and Collingridge (2015)	Macroecological model
2.4 Gt (40°N-40°S)	Anderson *et al.* (2019)	Food-web model
1.8 to 16.0 Gt (70°N -70°S)	Proud *et. al.* (2018)	Acoustics
[2.33;2.59] with mean 2.49 Gt (70°N – 70°S)	MICRORYS12V5	Macroecological model

Comparisons at regional levels are compiled based on published studies. Values from selected studies are shown (column "Range of biomass" in Table 7) and compared to corresponding biomass predicted by the model (last row of Table 7). These literature estimates exhibit a considerable variety of sampling and data processing approaches and results are often reported using different units. Particularly estimates based on net trawling have been corrected to account for capture efficiencies, which have been estimated to range from 14% (Koslow et al., 1997) to 20% (Davison et al., 2015) for mesopelagic fish, and which may also vary according to organism type/species (Ariza et al., 2015). Capture efficiency is likely higher for commercial-sized midwater trawls as in Hidaka et al. (2003); for these we applied an efficiency of 33%. These corrections need to be adapted for each study to account for the different methodologies used and depending on which layer was sampled during which part of the day. Most of the time, the epipelagic layer is targeted at night to measure migrating mesopelagic biomass. Therefore, the sum of corresponding functional groups in the model is used for the comparison.

The selected studies represent a diversity of oceanic regions from tropics to temperate waters and including coastal upwelling or oligotrophic areas. For all of them the model estimates compare relatively well with values ranging between 0.76 and 13.2 g WW m^{-2} for the model and 0.3-29.0 g WW for the observations (Table 7).

Table 7: Estimates of local mesopelagic fish biomass.

Region Period	Reference	Sampling	Method	Range of biomass (m^{-2})	Correction (gWW m^{-2})	Model average (gWW m^{-2})
California Current 2010-2012	Davison et al (2015)	35°N-30°N; 124°W-118°W	Acoustic model (18 and 38 kHz) with trawling Mesopelagic fish	25– 37 gWW	**12-28** (1)	**13.16** Average sum of all 5 mesopelagic groups
Northeast Atlantic (Canary) 24 Mar- 2 Jun 2011	Ariza et al (2015)	28°31'N; 15°22'W	Trawling acoustic (120 kHz) as indicator of vertical structure. All migrating organisms in the epipelagic layer	0.2 ±0.06 gC(2)	**2.4 ±0.7**	**2.79** Average sum in epipelagic layer at night
Western Tropical Pacific Mar -Apr 2010	Suntsov & Domokos (2013)	10°N-17°N 144°E-147°E	Trawling (IKMT) Night 0-200 m (All mnk) 0.8-1.2 mg m^{-3}	0.16-0.24 g WW (night 0-200 m)	**0.8-1.2** (3)	**0.87** Sum in epipelagic layer at night
North Atlantic (Azores) 20 Jun–6 Jul 2009	Cook et al (2013)	53°N 35°W	Trawling; data between 0-700 m only are selected. 0.009 g m^{-3} in 0-200 m 0.0045 g m^{-3} in 200-700 m	4.05 g WW (0-700 m)	**20** (3)	**3.70** All groups
Western tropical Pacific Nov-Dec 1998	Hidaka et al (2003)	2°N–20°N 125°–160°E	Trawling; commercial-sized midwater otter trawl 70 m long 0-200 m	St7: 0.212 St15: 0.388 St16: 0.192 gWW	**0.57 -1.16**(4)	**1.14** Average sum in epipelagic layer at night
East and South off New Zealand (Continental Plateau) Dec 2000-Jan 2001	McClatchie & Dunford (2003)	Mernoo & Snares East Campbell	Trawling and acoustic (38 kHz) Mesopelagic fish only	13.4-18.2 gWW 0.4 gWW 0.3 gWW	**13.4-18.2** **0.4** **0.3**	**15.58** **11.33** **7**
Tasman Sea 2004-07	Kloser et al (2009)	148°E-172°E; 41°S-45°S	Trawling and acoustic (38 and 120 kHz) Mesopelagic fish only	16-29 gWW	**16-29**	**12.79** Average of the sum of all 5 mesopelagic groups

(1) After removing 2 cruises likely biased by the presence of siphonophores.
(2) Correction from the authors for sampling efficiency: 14% for gas-bearing animals and 38% and 80% for large and small none-gas-bearing animals, respectively
(3) Correction for sampling efficiency: 20% (IKMT trawl)
(4) Correction for sampling efficiency: 33% (commercial-sized midwater otter trawl)

IV.3.2 Synthesis of product quality for micronekton

At both global and regional levels, model biomass estimates are consistent with estimates derived from other models and from either trawl sampling or combined acoustic and trawl sampling estimates using a conservative correction factor to account for trawl efficiency.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

Date	Change/Event description	System version	other
15/02/2019	1st release	version 1.2.03	
30/06/2023	Correction of the transport numerical scheme and change of the diffusion coefficient	version 4.1	
30/06/2024	Change of computation grid for Yin-Yang overset grids	version 4.2	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This new version (4.2) is based on a new version of the SEAPODYM model that uses a Yin-Yang overset computation grid (Kageyama *et. al. 2004)*.

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